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Modelling Disc Truncation in S-Type Binary Star Systems

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Abstract

Binary systems represent a significant fraction of stellar populations, with approximately half of Sun-like stars residing in multiple star systems. The truncation radius in S-type binary star systems, where planets orbit only one of the binary components, plays a critical role in the evolution and formation of circumstellar discs, directly influencing planet formation processes. The location of the truncation radius, which defines the disc's outer boundary, is dependent on parameters such as the mass ratio, binary separation, and binary eccentricity. In order to model the truncation radius, the method of Manara et al. (2019) is followed. This is done through Python code that first finds the mean value of the aspect ratio ($\frac{H}{r}$) from a population of discs around single stars, then interpolates the data from Table C.1 in Manara et al. (2019), and finally calculates the truncation radius. Using the first two steps of the Python code, the truncation radius is also calculated in Fortran. This is done so that it can be compatible with the Bern Model of Planet Formation which is written in Fortran, a comprehensive numerical model that simulates the growth and evolution of protoplanetary discs and planets of single-stars by accounting for accretion, migration, and dynamical interactions over time. This provides a comprehensive understanding of how binary star interactions shape protoplanetary discs, ultimately contributing to more realistic simulations of planet formation in binary systems.

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1 Introduction

1.1 Protoplanetary discs

Approximately half of Sun-like stars reside in multiple star systems (Raghavan et al. 2010). The most common type of these multiple star systems are binary star systems, which represent about 33% of all star systems (Raghavan et al. 2010). The planets found in binary star systems can be found in two types of orbits: S-type, where the planet orbits only one of the two stars in the binary, and P-type where the planet orbits both binary components (Cuntz 2014).

Planets are by-products of star formation. When a star forms from a rotating molecular cloud core, conservation of angular momentum forces the collapsing cloud matter to form a rotating accretion disc around the newly formed star, called a protostellar disc (Dullemond 2013). Turbulent viscosity drives much of the disc's material inward onto the star. After a few 10^5 years, most of the disc's mass ($M_{disc} << M_\star$) has accreted onto the star, the remaining material persists for about 10^7 years as an active accretion disc, providing sufficient time for planet formation. This late stage disc is called a protoplanetary disc (Dullemond 2013).

Circumstellar discs in multiple star systems are often more compact, with smaller dust disc radii compared to those around isolated stars (Manara et al. 2019). This means that either the discs are intrinsically smaller due to tidal truncation or that the dust grain drift is more efficient, driven by the truncated outer edge of the gas disc, accelerating dust growth and drift. Alternatively, smaller disc sizes in multiple systems may reflect smaller disc sizes at formation rather than being the result of evolution. Wide protostar binaries often form with large circumbinary discs ($\approx 300\ au$, (Takakuwa et al. 2017; Artur de la Villarmois et al. 2018)). Current research cannot yet determine if discs in multiple systems are small at the time of their formation.

There are two types of discs in a binary environment: circumstellar discs, which surround only one star and can lead to formation of S-type planets, and circumbinary discs, which surround the entire binary and can lead to formation of P-type planets. Two circumstellar discs, one around each star, and one circumbinary disc can be present in a binary at the same time (Artymowicz & Lubow 1994). However it is important to note that resolved circumbinary discs are not directly observed (Roddier et al. 1996). The focus of this work is on circumstellar discs, where the primary star's (the star with the larger mass) disc is called the circumprimary disc while the secondary star's (the star with the smaller mass) disc is called the circumsecondary disc.

For field stars, the typical binary star separation is about 30-50 au (Duquennoy & Mayor 1991), which is less than the typical T Tauri disc size of a few hundred au. Consequently, binary star systems will usually interact strongly with discs (Beckwith et al. 1990). These interactions result in tidal truncation of the disc, governed by factors such as binary separation, mass ratio, eccentricity, and the overall dynamics of the system.

1.2 Viscous disc profile

If the gas in the Keplerian disc is perfectly non-viscous, each fluid element has angular momentum $l = \Omega_K r^2$ where Ω_K is the Keplerian angular frequency:

$$\Omega_K = \sqrt{\frac{GM_\star}{r^3}}$$

where G is the gravitational constant and $M_\star$ the stellar mass. Resulting in $l \propto \sqrt{r}$. For accretion to occur, l must decrease as r decreases. Without torque, fluid elements remain in circular orbits without accreting. Viscosity can, however, lead to an outward redistribution of angular momentum: the inner disc transports angular momentum to the outer disc, so

that the inner disc moves inward, while the outer disc moves outward (Dullemond 2013). The viscosity generated by molecular collisions (ν_{mol}) in accretion discs is extremely low, making accretion timescales billions of years. However, turbulence, driven by mechanisms such as magnetorotational instability (MRI) or baroclinic instability (BI), creates effective turbulent viscosity. This is often modeled using the α-viscosity prescription:

$$\nu = \alpha \frac{c_s^2}{\Omega_K} \tag{1}$$

or in terms of the mean free path λ modelled with the Navier-Stokes equations:

$$\nu_m \approx \lambda c_s \tag{2}$$

where

$$c_s = \sqrt{\frac{k_B T}{\mu m_p}}$$

is the isothermal sound speed, where k_B is the Boltzmann constant, μ is the mean molecular weight in units of the proton mass m_p ($\mu = 2.3$ for a fully molecular gas of cosmic composition). The dimensionless α-parameter, known as the Shakura-Sunyaev α parameter, measures the efficiency of angular momentum transport due to turbulence (Shakura & Sunyaev 1973). It must be $\alpha \lesssim 1$ and it is typically assumed to lie within the range $[10^{-3}, 10^{-2}]$. However, in this lab, we focus on the range $[10^{-4}, 10^{-3}]$. We will take α to be constant because, in global formation models like the Bern Model of Planet Formation—a comprehensive numerical model that simulates the growth and evolution of protoplanetary discs and planets—α is treated as constant to reduce the computational cost.

The consequence of a small molecular viscosity is a large Reynolds number which is defined as:

$$\mathcal{R} = \frac{UL}{\nu_m}$$

where U and L are characteristic velocity and length scales in the system. The presence of a physical instability means that the protoplanetary disc will be highly turbulent, with dissipation of fluid motions occurring on a scale that is small compared to the disc scale-height (Armitage 2010). The turbulence could be due to gravitational instability (Lin & Pringle 1987) or magnetic shearing instability (Balbus & Hawley 1991). Due to these complexities, modelling the disc with a simple viscosity is an approximation (Artymowicz & Lubow 1994).

Like with the radial structure, the vertical density structure is related to the vertical temperature structure. Assuming the temperature $T(r, z)$ is independent of z ($\frac{\partial T}{\partial z} = 0$) results in the good approximation $\frac{\partial c_s^2}{\partial z} = 0$ (Dullemond 2013). The vertical hydrostatic equilibrium equation in the disc is therefore:

$$\frac{\partial p}{\partial z} = -\rho \frac{GM_*}{r^3} z \equiv -\rho \Omega_K^2 z$$

with $p = \rho c_s^2$ and with $\frac{\partial c_s^2}{\partial z} = 0$ this becomes:

$$\frac{1}{\rho}\frac{\partial \rho}{\partial z} = -\frac{\Omega_K^2}{c_s^2} z$$

whose solution is:

$$\rho(z) = \rho_0 e^{-\frac{z^2}{2H^2}}$$

Where we define the pressure scale height or vertical thickness H as:

$$H = \frac{c_s}{\Omega_K} \tag{3}$$

If we have the typical temperature profile for irradiated discs of $T \propto r^{-1/2}$ then $c_s \propto r^{-1/4}$ so that:

$$H \propto r^{-1/4+6/4} \propto r^{5/4}$$

This means $\frac{H}{r} \propto r^{1/4}$, i.e. it increases with r. This results in a disc with a flaring geometric shape (Dullemond 2013).

The disc's large surface area allows efficient radiative cooling, resulting in relatively low temperatures and pressures. This prevents gas from resisting gravity except in a geometrically thin disc configuration. As a result, the vertical thickness H is much smaller than the orbital radius r resulting in $\frac{H}{r} << 1$ (Armitage 2010).

1.3 Effects of a binary on discs

The tidal forces caused by the binary generally distort discs from their circular shape. Since most binaries are eccentric, they cause time-dependent deformations at the binary frequency and its harmonics. At resonance points, strong interactions between the disc and binary often occur. These resonant interactions usually dominate (Lubow & Artymowicz 2000).

Goldreich & Tremaine (1980) developed the theory of disc resonances, allowing analytical expressions for torques exerted at these resonances to be obtained. To do this, the binary potential Φ can be decomposed into Fourier components:

$$\Phi(r, \theta, t) = \sum_{m,l} \phi_{ml}(r) cos(m\theta - l\Omega_b t)$$

or:

$$\Phi(r, \theta, t) = \sum_{m,l} \phi_{ml}(r) exp[i(m\theta - l\Omega_b t)]$$

where the cylindrical coordinates (r, θ) are centred on the binary centre of mass in the inertial frame, and Ω_b is the binary's angular frequency. For circular binaries, only diagonal terms (m=l) are non-zero, making the potential static in the binary's frame. Eccentric binaries (typically the case) introduce non-diagonal terms (l $\neq$ m), in this case $|\Phi|$ scales with the eccentricity as $e^{|m-l|}$ (Lubow & Artymowicz 2000).
Waves that exert torques as they damp are released at two types of resonances:

1. Lindblad resonances occur where $\Omega(r) = \frac{l\Omega_b}{m \pm 1}$. They truncate discs when $m = l$ or truncate discs and increase eccentricity when $m \neq l$ (eccentric Lindbland resonances).

2. Corotational resonances occur where $\Omega(r) = \frac{l\Omega_b}{m}$. They generally damp binary eccentricity.

Whilst it is important to keep these resonant interactions in mind, they were not the focus of this work, so they were not accounted for in the calculations.

2 Method

2.1 Theoretical prescription

Manara et al. (2019) derived an analytical function for the truncation radius of circumstellar discs in binary systems, as predicted by the theory of Artymowicz & Lubow (1994). Artymowicz & Lubow (1994) determined the disc truncation by balancing resonant torques with disc viscous torques, establishing its dependence on the mass ratio, orbital eccentricity, and Reynolds number of the disc, defined as $\mathcal{R} = \frac{1}{\alpha}(\frac{r}{H})^2$, assuming the disc and binary orbit are co-planar.

When the orbital eccentricity e is zero, disc truncation is dominated by non-resonant interactions (Papaloizou & Pringle 1977). In this case, the truncation radius depends only on the mass ratio and not on viscosity, and it can be expressed in terms of the mass parameter $\mu = \frac{M_2}{M_1+M_2}$, with M_1 and M_2 being the masses of the primary and secondary stars, respectively.

Manara et al. (2019) fitted the truncation radius as a function of both e and μ using an exponential expression, which is multiplied by the Roche lobe radius ($R_{i,Egg}$), the radius of the region around a star in a binary system within which orbiting material is gravitationally bound to that star (Eggleton 1983), of the respective star. The term of μ is small compared to R_{i_Egg}, so most of the dependence on the masses will be inside R_{i_Egg}. The fitting function is given by:

$$R_t(M_1, M_2, e, a_{bin}) = R_{i,Egg}(be^c + h\mu^k) \tag{4}$$

where b, c, h, and k are the fitting parameters and

$$\frac{R_{i,Egg}}{a_{bin}} = \frac{0.49q_i^{2/3}}{0.6q_i^{2/3} + ln(1 + q_i^{1/3})} \tag{5}$$

where $q_1 = \frac{M_1}{M_2}$, $q_2 = \frac{M_2}{M_1}$ and a_{bin} is the binary separation. We consider $q = q_2$ to be the mass ratio.

The first term of the Equation (4) contains all of the dependence on the eccentricity, while the second term describes how the zero-eccentricity truncation radius changes with varying (M_1, M_2, a_{bin}).

Manara et al. (2019) determined the fitting parameters h and k by fitting the results obtained by Papaloizou & Pringle (1977), yielding $h = 0.88$ and $k = 0.01$ (both data from Papaloizou & Pringle (1977) and fits are illustrated in Figure 1). The dependence on the eccentricity (i.e. b and c) was found by Manara et al. (2019) by fitting the numerical results from Artymowicz & Lubow (1994) with

$$R_t(M_1, M_2, e, a) = R_{i,Egg}(be^c + 0.88\mu^{0.01}) \tag{6}$$

Table 1 reports the fitting parameters for a range of mass parameters μ and Reynolds numbers for both the circumprimary and the circumsecondary disc. The fitting parameters show minor variations with μ, remaining relatively constant across its range.

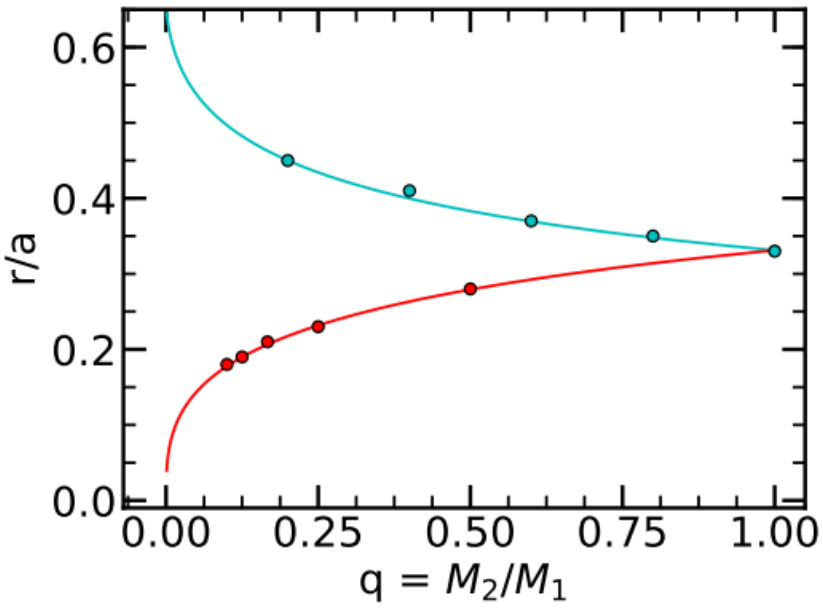


Figure 1: Tidal truncation radius predictions for a circular binary (i.e., $e = 0$) as a function of the mass ratio q (taken from Fig. C.1. of Manara et al. (2019)). The solid lines represent the fitting function $R_{i,Egg}0.88\mu^{0.01}$. The blue line refers to the circumprimary disc while the red line to the circumsecondary. The blue and red dots are the theoretical predictions of Papaloizou & Pringle (1977) for the circumprimary and circumsecondary disc, respectively.

Circumprimary			Circumsecondary		
$\mathcal{R}$	b	c	$\mathcal{R}$	b	c
$\mu = 0.1$					
10^4	-0.66	0.84	10^4	-0.81	0.98
10^5	-0.75	0.68	10^5	-0.81	0.80
10^6	-0.78	0.56	10^6	-0.83	0.69
$\mu = 0.2$					
10^4	-0.72	0.88	10^4	-0.81	0.99
10^5	-0.78	0.72	10^5	-0.82	0.82
10^6	-0.80	0.60	10^6	-0.83	0.70
$\mu = 0.3$					
10^4	-0.76	0.92	10^4	-0.79	0.97
10^5	-0.80	0.75	10^5	-0.82	0.81
10^6	-0.81	0.63	10^6	-0.83	0.69
$\mu = 0.4$					
10^4	-0.77	0.95	10^4	-0.80	0.98
10^5	-0.81	0.78	10^5	-0.82	0.80
10^6	-0.82	0.66	10^6	-0.83	0.68
$\mu = 0.5$					
10^4	-0.78	0.94	10^4	-0.79	0.95
10^5	-0.81	0.78	10^5	-0.81	0.78
10^6	-0.82	0.66	10^6	-0.82	0.66

Table 1: Best fit parameters of Equation (6) for different values of μ and $\mathcal{R}$, for both the circumprimary and circumsecondary discs (Table C.1 from Manara et al. (2019)).

2.2 Implementation

Implementing the truncation radius formula reported in Equation (6) involved a multistep process. Out of the four parameters in Equation (6) (i.e., b, e, c, μ), e is given as an input parameter while μ is determined from M_1 and M_2 (which are also given as input parameters). Therefore, only the values of b and c must be found.

In order to do this, using the values in Table 1, for each μ a quadratic fit was done for both b and c as a function of the Reynolds number, $\mathcal{R}$, for both the cirumprimary and circumsecondary cases, resulting in a total of 20 polynomials. These polynomials were used to determine the values of b and c as a function of μ and $\mathcal{R}$, which finally gives all the required parameters to calculate the truncation radius using Equation (6).
The quadratic fits of $b(\mathcal{R})$ and $c(\mathcal{R})$ for the circumprimary disc are given by:

$$
\begin{aligned}
b_1 &= 9.764309764309791 \cdot 10^{-13} R^2 - 1.1074074074074087 \cdot 10^{-6} R - 0.6490235690235693 \\
c_1 &= 1.6611 \cdot 10^{-12} R^2 - 1.9605 \cdot 10^{-6} R + 0.85944 \\
b_2 &= 6.509539842873163 \cdot 10^{-13} R^2 - 7.382716049382692 \cdot 10^{-7} R - 0.7126823793490461 \\
c_2 &= 1.6611 \cdot 10^{-12} R^2 - 1.9605 \cdot 10^{-6} R + 0.89944 \\
b_3 &= 4.3771043771043816 \cdot 10^{-13} R^2 - 4.925925925925918 \cdot 10^{-7} R - 0.7551178451178455 \\
c_3 &= 1.7733 \cdot 10^{-12} R^2 - 2.0840 \cdot 10^{-6} R + 0.94066 \\
b_4 &= 4.3771043771043937 \cdot 10^{-13} R^2 - 4.925925925925932 \cdot 10^{-7} R - 0.7651178451178456 \\
c_4 &= 1.7733 \cdot 10^{-12} R^2 - 2.0840 \cdot 10^{-6} R + 0.97066 \\
b_5 &= 3.2547699214366364 \cdot 10^{-13} R^2 - 3.691358024691398 \cdot 10^{-7} R - 0.7763411896745228 \\
c_5 &= 1.6611 \cdot 10^{-12} R^2 - 1.9605 \cdot 10^{-6} R + 0.95944
\end{aligned}
$$

While for the circumsecondary disc they are given by:

$$
\begin{aligned}
b_1 &= -2.244668911335608 \cdot 10^{-14} R^2 + 2.4691358024707388 \cdot 10^{-9} R - 0.8100224466891136 \\
c_1 &= 1.8967 \cdot 10^{-12} R^2 - 2.2086 \cdot 10^{-6} R + 1.0019 \\
b_2 &= 1.0101010101009552 \cdot 10^{-13} R^2 - 1.2222222222221502 \cdot 10^{-7} R - 0.808787878787879 \\
c_2 &= 1.7733 \cdot 10^{-12} R^2 - 2.0840 \cdot 10^{-6} R + 1.0107 \\
b_3 &= 3.254769921436567 \cdot 10^{-13} R^2 - 3.691358024691327 \cdot 10^{-7} R - 0.7863411896745232 \\
c_3 &= 1.6611 \cdot 10^{-12} R^2 - 1.9605 \cdot 10^{-6} R + 0.98944, \\
b_4 &= 2.132435465768843 \cdot 10^{-13} R^2 - 2.456790123456824 \cdot 10^{-7} R - 0.7975645342312008 \\
c_4 &= 1.8855 \cdot 10^{-12} R^2 - 2.2074 \cdot 10^{-6} R + 1.0019 \\
b_5 &= 2.1324354657688054 \cdot 10^{-13} R^2 - 2.456790123456777 \cdot 10^{-7} R - 0.7875645342312014 \\
c_5 &= 1.7733 \cdot 10^{-12} R^2 - 2.0840 \cdot 10^{-6} R + 0.97066
\end{aligned}
$$

These 20 polynomials can be better represented by plotting the fits. Figures 2 and 3 show the polynomial fits for $b(\mathcal{R})$ and $c(\mathcal{R})$ as a function of the Reynolds number $\mathcal{R}$. Each figure contains two graphs (one for $b(\mathcal{R})$ and one for $c(\mathcal{R})$), with each displaying the five fits for the five values of μ (0.1, 0.2, 0.3, 0.4, 0.5). The scatter points represent the data from Table 1.

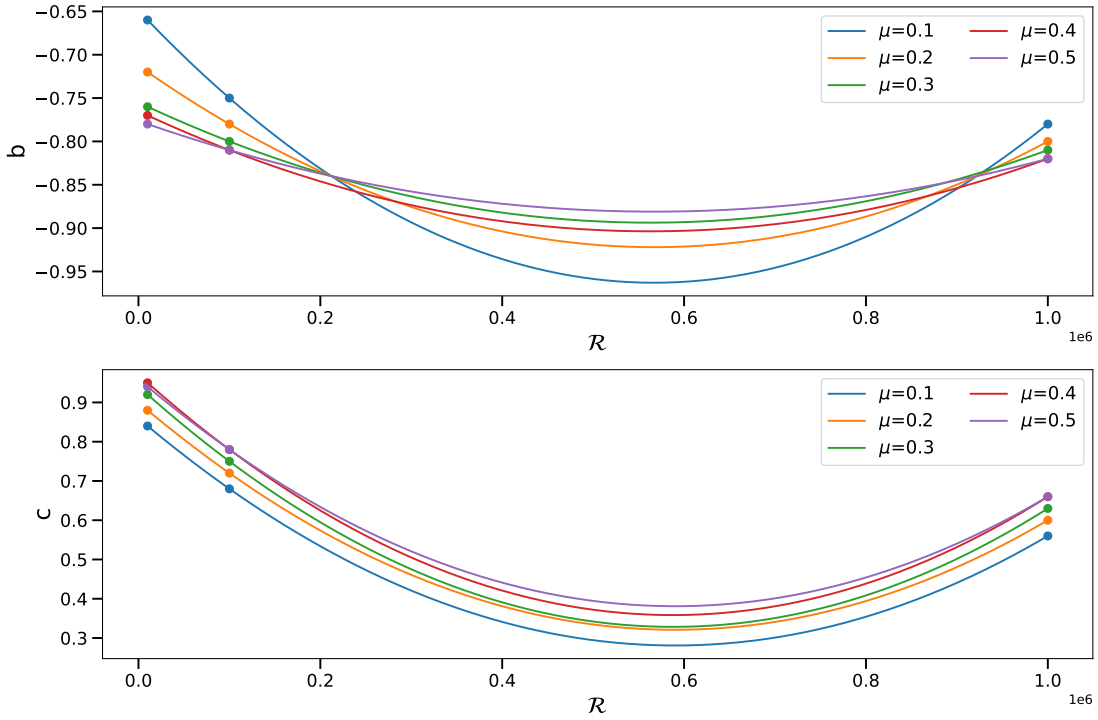


Figure 2: Circumprimary quadratic fits of $b(\mathcal{R})$ (top) and $c(\mathcal{R})$ (bottom) for each value of μ using the data of Table 1. With μ = [0.1, 0.2, 0.3, 0.4, 0.5] being represented by the colours [blue, orange, green, red, purple] respectively.

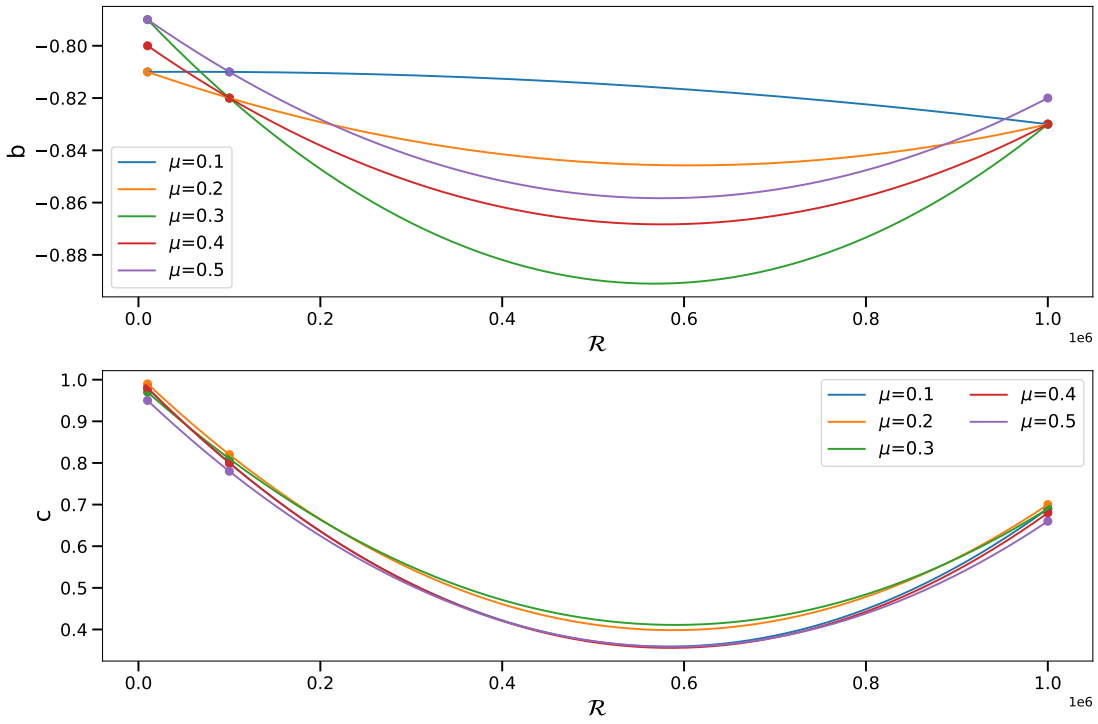


Figure 3: Circumsecondary quadratic fits of $b(\mathcal{R})$ (top) and $c(\mathcal{R})$ (bottom) for each value of μ using the data of Table 1. With μ = [0.1, 0.2, 0.3, 0.4, 0.5] being represented by the colours [blue, orange, green, red, purple] respectively.

In order to calculate the Reynolds number, which can then be used in the fits mentioned above, the value of $\frac{r}{H}$ needs to be found. Using the results from a population (i.e., 1000 simulations, run using the Bern Model of Planet Formation (Emsenhuber et al. 2021)) of protoplanetary discs around single stars, we obtain 1000 mean values for the disc aspect ratio $\frac{H}{r}$, one for each simulated disc. These simulations were conducted for 3 different values of the Shakura-Sunyaev α parameter: 10^{-3}, 10^{-4}, and $5 \cdot 10^{-4}$. The mean of $\frac{H}{r}$ is calculated in 3 different ways: across the full disc, for orbital distances greater than 10 au, and for orbital distances less than 10 au.

Looking at the initial (t = 0 years) $\frac{H}{r}$ as a function of the orbital distance, for viscosity of 10^{-3} and viscosity of 10^{-4}, we see that after an orbital distance of 10 au the values of $\frac{H}{r}$ converge. Therefore, to simplify the problem, one can take the mean of $\frac{H}{r}$ for orbital distances greater than 10 au to get a constant value, regardless of the choice of α. If instead the mean is taken for all orbital distances or only for orbital distances less than 10 au, it would depend on the value of α. In this case, a polynomial fit is done to best approximate $\frac{H}{r}$ based on the given α.

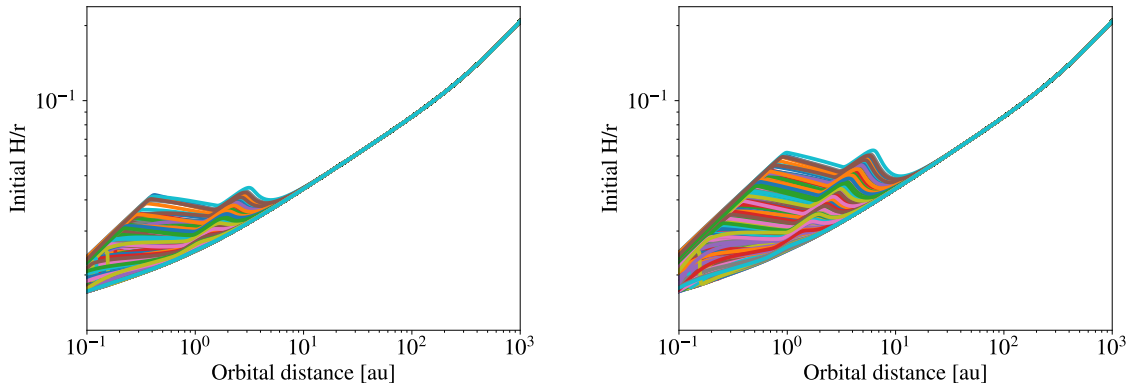


Figure 4: $\frac{H}{r}$ for the 1000 simulations as a function of orbital distance. In the left panel we show $\alpha = 10^{-4}$ while, in the right panel we show $\alpha = 10^{-3}$.

To estimate the aspect ratio for a given α and disc interval, a Gaussian distribution was fitted to each set of 1000 values. The mean of this Gaussian represents the best estimate for $\frac{H}{r}$.

As shown on the top panel of Figure 5, the mean values of $\frac{H}{r}$ differ. This variation is expected as orbital distances less than 10 au are considered, which we know from Figure 4 that the values of $\frac{H}{r}$ do not converge at these distances. This is even more pronounced on the bottom panel of Figure 5, where there is greater variation in the means of the Gaussian fits because $\frac{H}{r}$ is averaged from the inner edge up to 10 au only, revealing more differences in the $\frac{H}{r}$ profile.

The 3 values found on Figure 6 are the same up to 17 decimal points (which are all that Python gives). This value is $\frac{H}{r} = 0.09826384282379806$. While the polynomial fit using the values of the top panel of Figure 5 is $\frac{H}{r}(\alpha) = 6.037912 \cdot 10^{8}\alpha^{2} - 9.439091 \cdot 10^{6}\alpha + 3.268866 \cdot 10^{4}$.

Having found an estimate for $\frac{H}{r}$, the value of the Reynolds number $\mathcal{R}$ can be calculated. This in turn gives the values of $b(\mathcal{R})$ and $c(\mathcal{R})$ and, therefore, all the parameters needed for the implementation of Equation (6).

Since averaging $\frac{H}{r}$ over the outer part of the disc gives a constant value, simplifying the calculations, in the following analysis we consider this specific average of $\frac{H}{r}$.

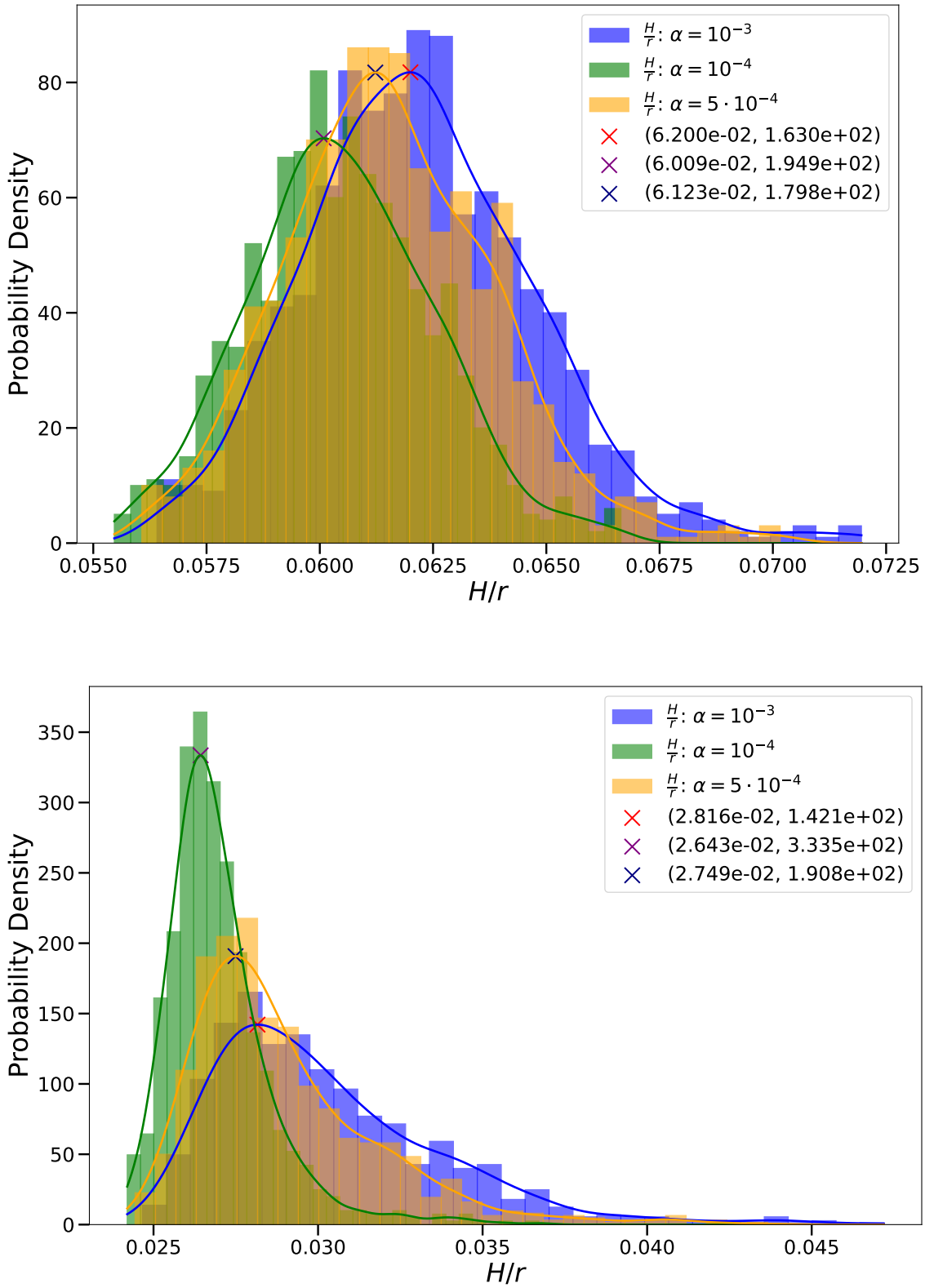


Figure 5: Probability density functions and histograms of the $\frac{H}{r}$ for $\alpha = [10^{-3}, 10^{-4}, 5 \cdot 10^{-4}]$ represented by the colours [blue, green, orange] respectively. The X marks represent the means of each PDF with the colours [red, purple, navy] corresponding to the same order of α. The top panel shows the $\frac{H}{r}$ values averaged over the full disc. The bottom panel shows the $\frac{H}{r}$ values averaged between the inner disc edge and 10 au.

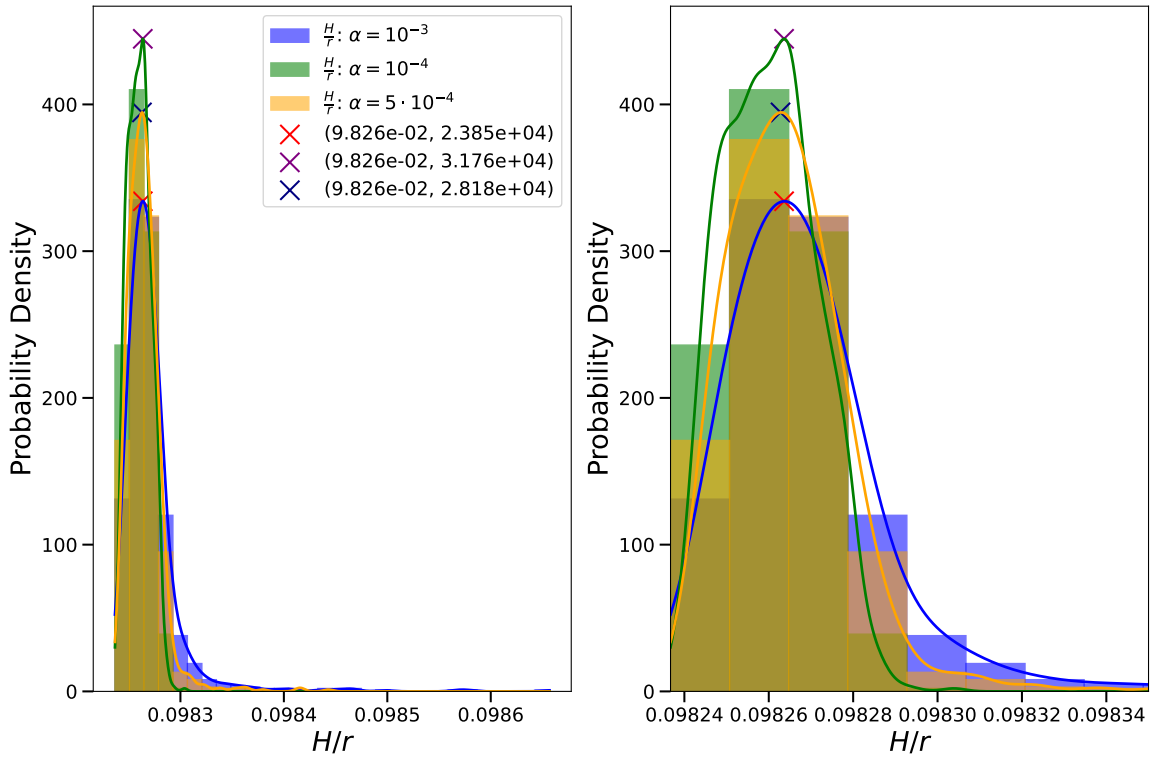


Figure 6: Probability density functions and histograms of the $\frac{H}{r}$ averaged for orbital distances greater than 10 au for $\alpha = [10^{-3}, 10^{-4}, 5{\cdot}10^{-4}]$ represented by the colours [blue, green, orange] respectively. The X marks represent the means of each PDF with the colours [red, purple, navy] corresponding to the same order of α. The left panel shows the regular plot while the right panel, since the regular plot is characteristically Poissonian, shows a zoomed in version where the maximum value of the x-axis is 0.09835.

3 Results

3.1 Analysis

To test whether the implementation of Equation (6) was done correctly, we recreate Figure 1. The points on the figure are data from Papaloizou & Pringle (1977) while the solid lines are the fitting functions done by Manara et al. (2019). The data is retaken and a fit is done for 0 eccentricity so that like in Figure 1 we only measure the dependence on the masses.

The resulting plot correctly fits the data from Papaloizou & Pringle (1977), which means that the implementation was done successfully. We see that as $M_2 \to 0$, the secondary star's gravitational influence on the circumprimary disc diminishes rapidly, allowing the cirumprimary truncation radius to expand. Simultaneously, the circumsecondary truncation radius decreases at the same rate as the circumprimary's growth, as the secondary star loses its ability to retain material. Conversely, when $q = 1$, the symmetry of the gravitational forces ensures that the two truncation radii converge to the same value.

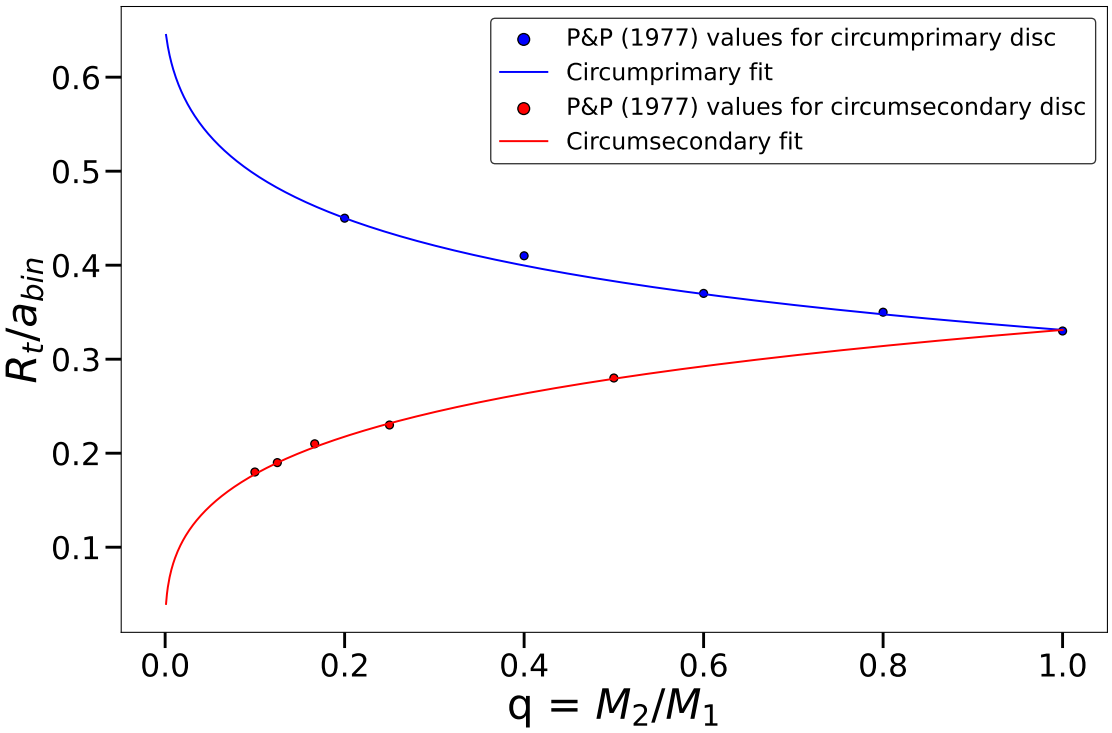


Figure 7: Recreation of Figure 1 done by creating fits (the solid lines) for the data from Papaloizou & Pringle (1977) (the circle markers). The circumprimary disc is in blue while the circumsecondary disc is in red.

In Figure 7 only the dependence of the truncation radius on the masses is measured. However, the influence of the eccentricity can also be measured as a function of $\mathcal{R}$, provided that μ and e are fixed. In this case, the eccentricity is set to 0.5, while the same values of μ as in Table 1 are used. Figure 8 shows that $b \cdot e^c$ is inversely proportional to $\mathcal{R}$. Specifically, as $\mathcal{R}$ increases, the absolute value of $b \cdot e^c$ increases. As the Reynolds number increases, the system's viscosity decreases, which reduces the effect of viscous forces. With lower viscosity, the material in the disc is less damped and becomes more sensitive to other forces such as pressure and gravitational forces. As a result, the disc's ability to resist compression decreases, leading to a smaller truncation radius. Due to the minor variations in the fitting parameters b and c with μ, no clear trend in the dependence of $b \cdot e^c$ on μ can be discerned from the graph.

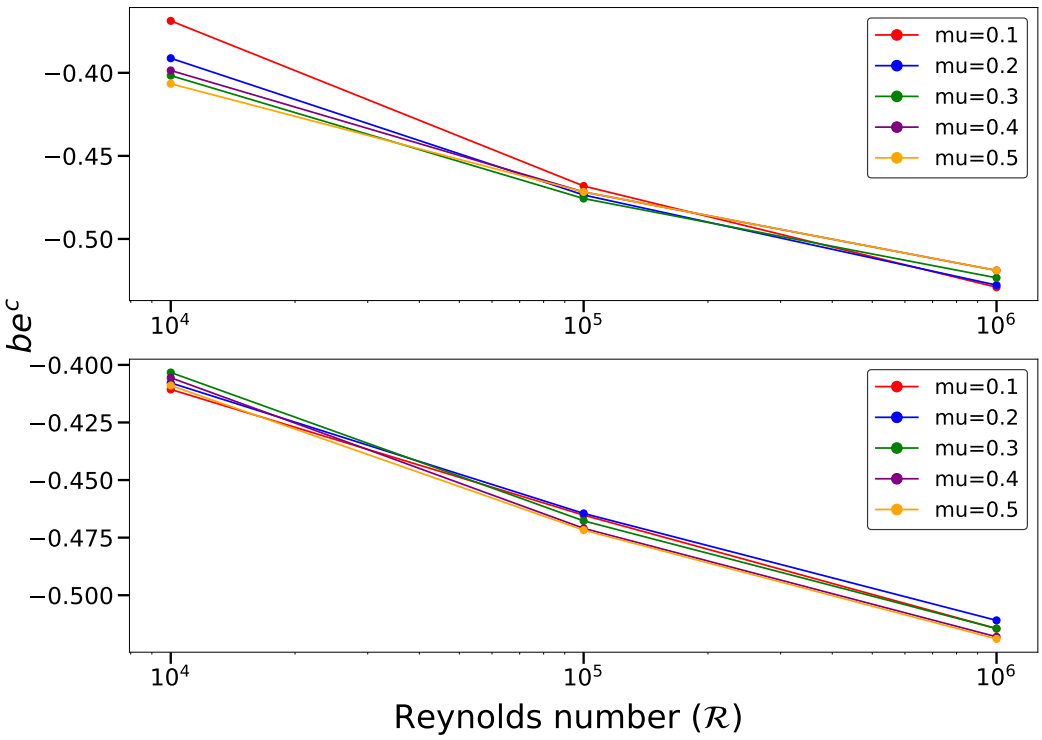


Figure 8: be^c as a function of $\mathcal{R}$ for fixed different fixed μ when $e = 0.5$. The x-axis uses a log scale to better visualise the data. With $\mu = [0.1, 0.2, 0.3, 0.4, 0.5]$ represented by the colours [red, blue, green, purple, orange] respectively. The plots for the circumprimary disc are on top while the plots for the circumsecondary disc are on the bottom.

We can also see the effects of the variables in Equation (6) on the truncation radius individually. To see the dependence on an individual variable, the rest need to be fixed. In this case, (when not the dependent variable) we fix $q = 0.5$, $\mu = \frac{1}{3}$, $e = 0.5$, and $\alpha = 10^{-3}$. In Figure 9, we see that the top two panels show the same trends as Figure 7 and Figure 8. $R_t(q)/a_{bin}$ of the circumprimary and circumsecondary converge to the same value at $q = 1$ and the circumprimary/circumsecondary grows/decays rapidly as $q \to 0$. We also see again that for both discs $R_t(e)/a_{bin}$ decreases with eccentricity.

With $R_t(\alpha)/a_{bin}$ we see that at low α, gravitational forces dominate, while at higher α viscosity redistributes angular momentum, changing the truncation radius dynamics by enhancing outward and inward migration of material. The minima represent the points where the balance of forces shifts. At this point the tidal forces from the opposing star overcome the viscous forces within the disc, leading to a maximum inward truncation of the disc. Since the circumprimary experiences less tidal forces due to the secondary star's mass being smaller, it is less constrained to drive material inward. As α increases beyond the minima viscosity dominates and drives outward motion overcoming gravitational and tidal forces.

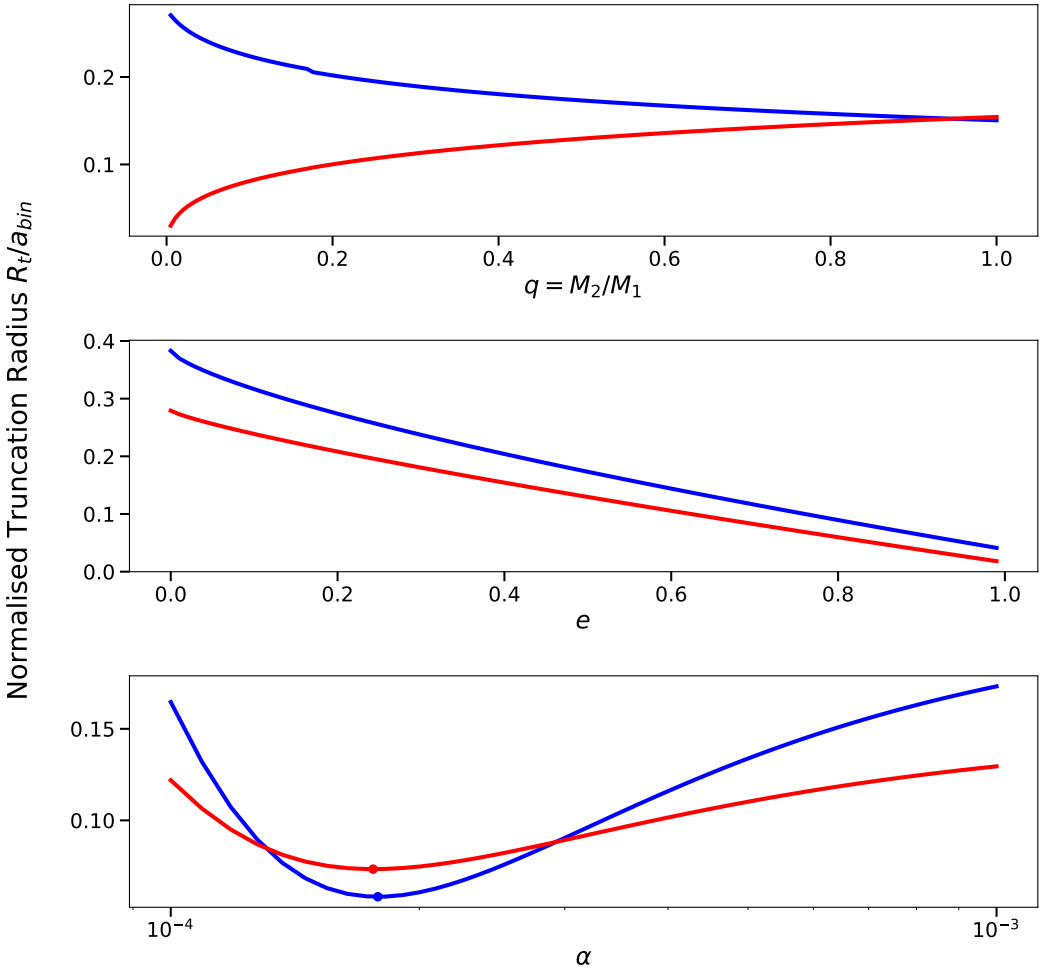


Figure 9: R_t/a_{bin} as a function of q (top), as a function of e (middle), and as a function of α (bottom). The circumprimary disc is represented in blue while the circumsecondary disc is represented in red. In the bottom graph, $R_t(\alpha)/a_{bin}$, the red minima is at $(1.76 \times 10^{-4}, 0.07)$ while the blue minima is at $(1.78 \times 10^{-4}, 0.06)$.

These relationships can be further looked into by considering the multivariable case. In Figure 10 we see again the relationship between decreasing R_t as both q and e increase. We also see a downward shift of the curves at increasing α (the minima) followed by an upward shift for large α. In Figure 11 there is the same relationship with a decrease in R_t as q increases as well as a decrease in R_t as a_{bin} increases as the stars' gravitational effects on each other decreases. In Figure 11 there is also a downward shift as e gets larger. In Figure 12 we see the same relationship with α as in Figure 9. Except when $e = 0$ as it is just constant as a function of α since the dependence on the viscosity is inside the term $b \cdot e^c$.

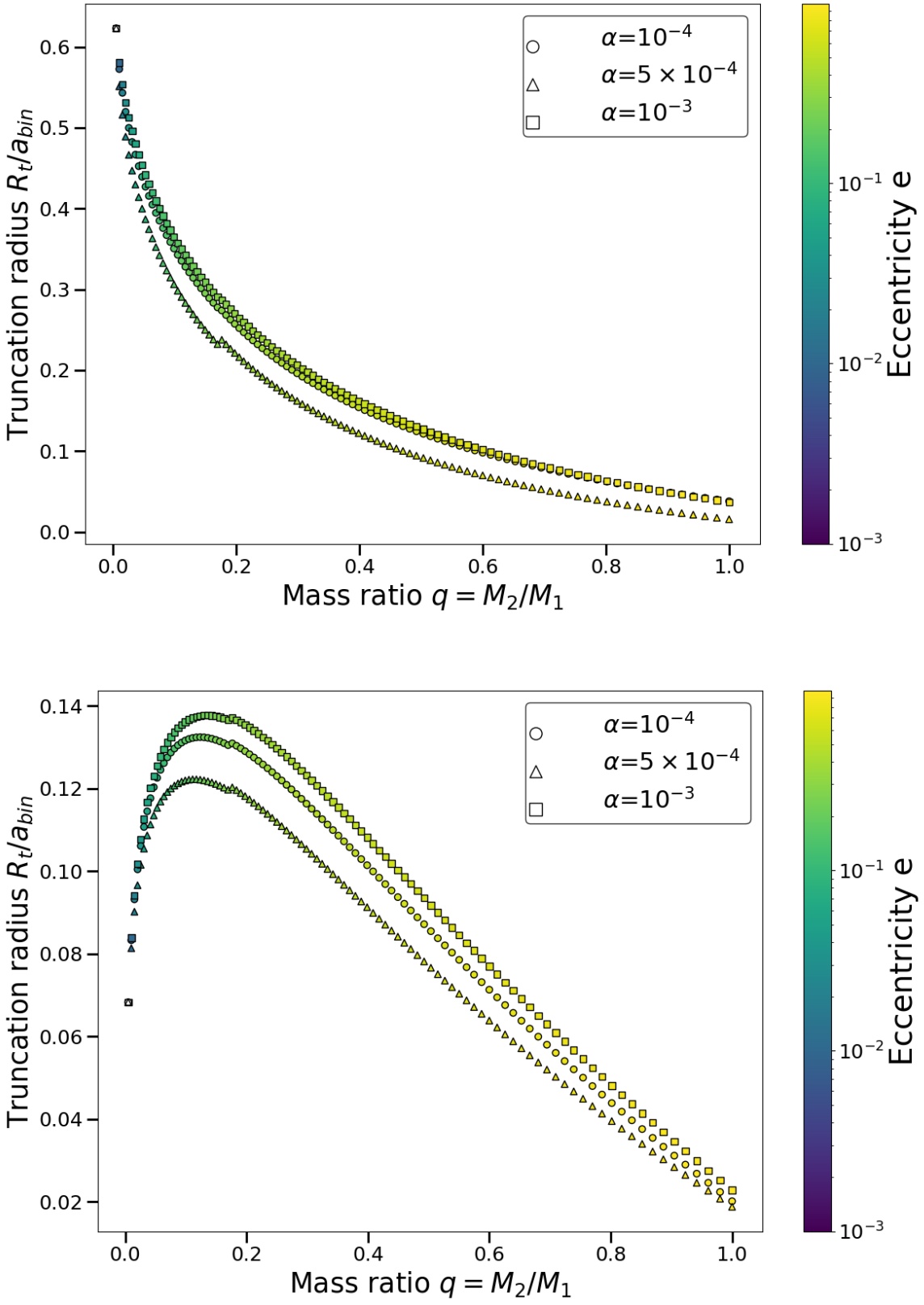


Figure 10: $R_t(q,e)/a_{bin}$ with different fixed α. The values of $\alpha = [10^{-4}, 5 \cdot 10^{-4}, 10^{-3}]$ correspond to the shapes [circle, triangle, square] respectively. The top panel refers to the circumprimary disc, while the bottom panel to the cirucmsecondary disc.

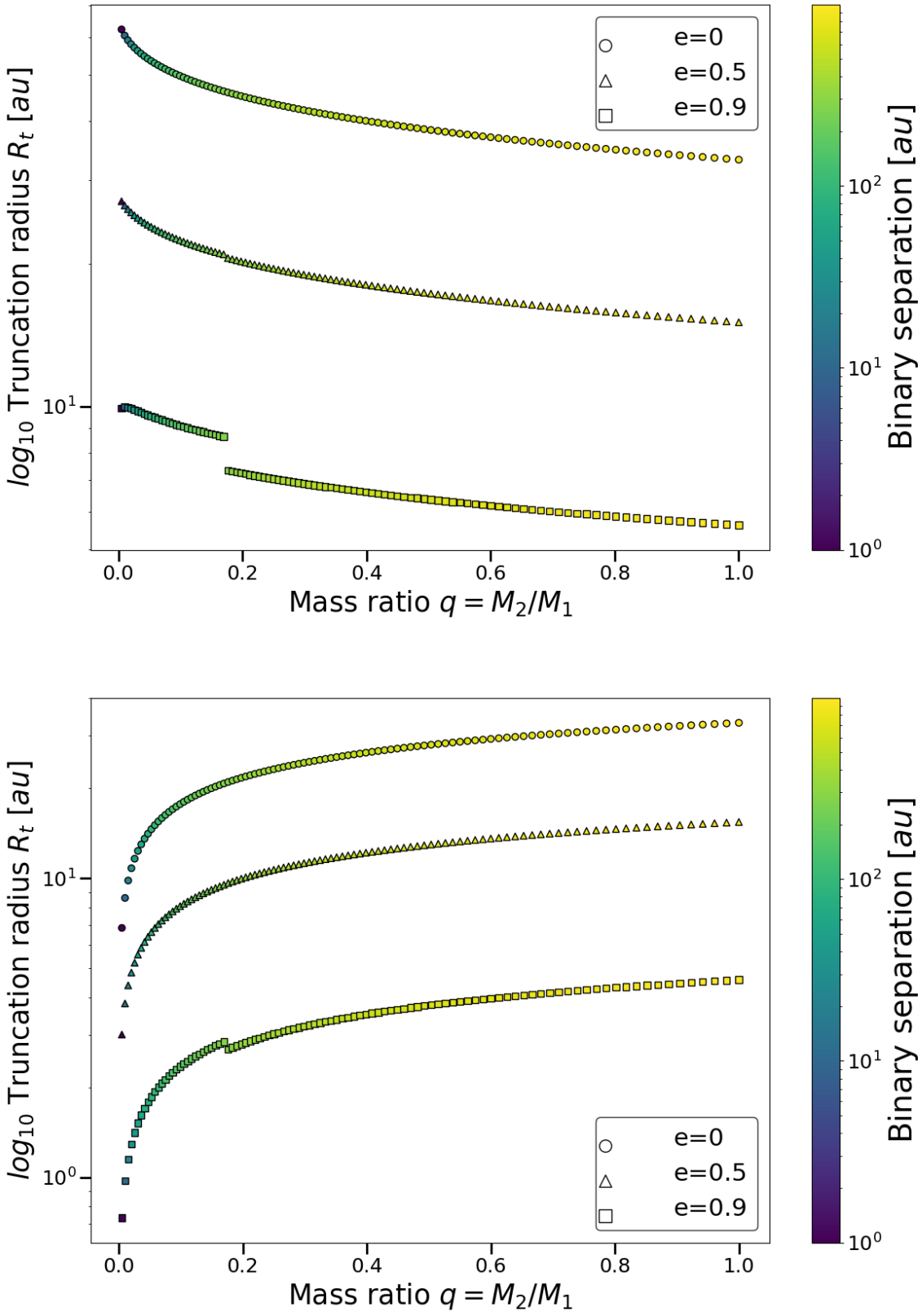


Figure 11: $log_{10}[R_t(q, a_{bin})]$ with different fixed e. The values of e = [0, 0.5, 0.9] correspond to the shapes [circle, triangle, square] respectively. The top panel refers to the circumprimary disc, while the bottom panel to the cirucmsecondary disc.

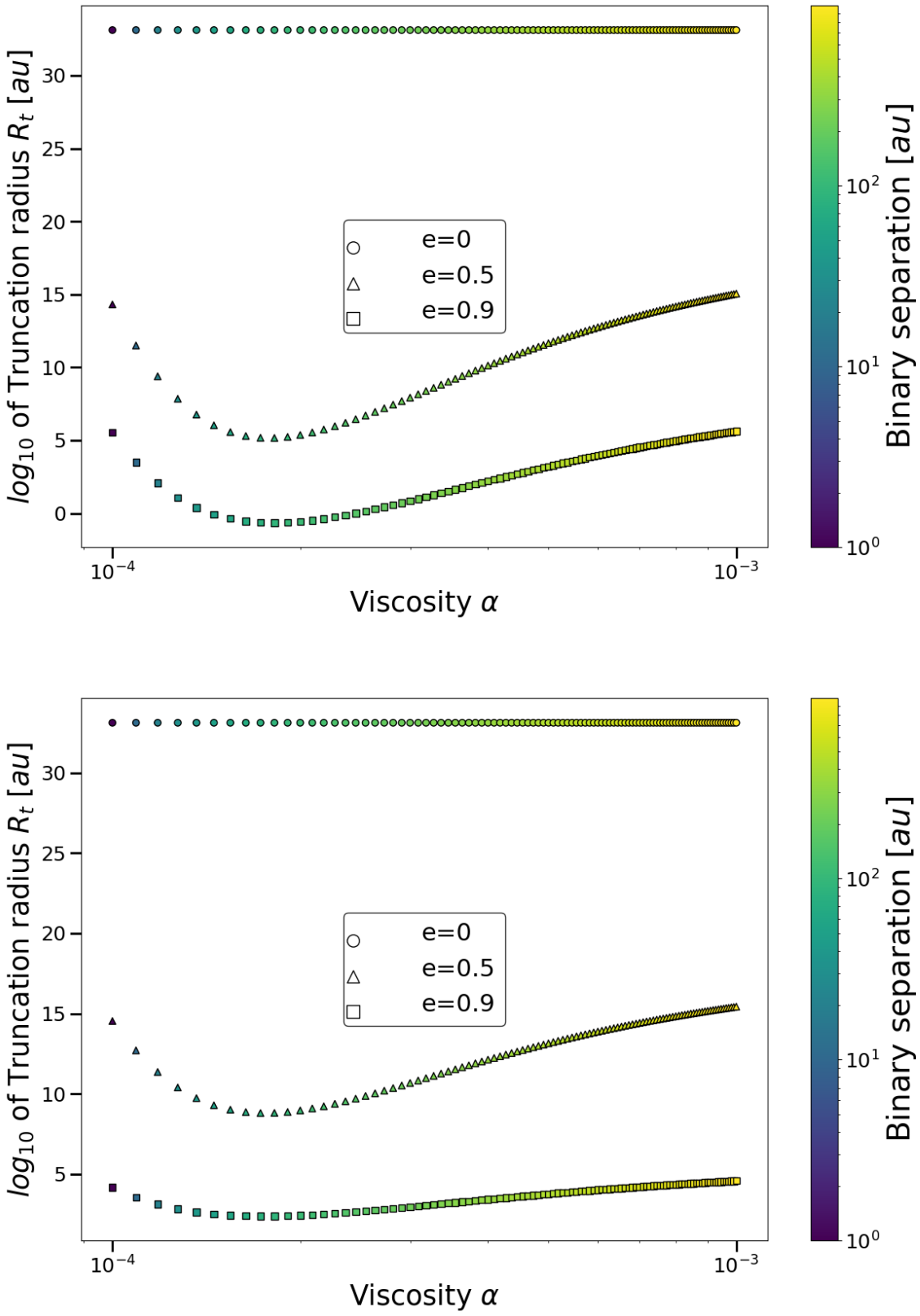


Figure 12: $log_{10}[R_t(\alpha, a_{bin})]$ with different fixed e and fixed q. The x-axis is in the log scale. The values of e = [0, 0.5, 0.9] correspond to the shapes [circle, triangle, square] respectively. The top panel refers to the circumprimary disc, while the bottom panel to the cirucmsecondary disc.

3.2 Implementation in the Fortran language

In the previous section, the results were obtained using the implementation of the truncation radius formula in the Python language. However, the calculation of the truncation radius was also implemented in Fortran. This was necessary because the Bern model is written in Fortran, and if the truncation radius calculation is to be integrated into it, then it must be Fortran. Additionally, Python is not well-suited for computationally intensive tasks, while Fortran excels in such scenarios, often surpassing even C++ in performance.

Using the values $M_1 = 1$, $M_2 = 0.5$, $a_{\text{bin}} = 100\ au$, $e = 0.5$, $\alpha = 10^{-3}$ and using the $\frac{H}{r}$ averaged across the outer part of the disc, as done in the previous sections, we can compare the values given by Fortran and Python given the same initial conditions.

Disc Type	Language	Value	Difference
Circumprimary	Fortran	17.319933	3.5×10^{-7}
	Python	17.319932	
Circumsecondary	Fortran	12.9510332	2.8×10^{-7}
	Python	12.9510335	

Table 2: Comparison of truncation radius values between Fortran and Python for circumprimary and circumsecondary discs using $\frac{H}{r}$ averaged across the outer part of the disc.

Using instead the $\frac{H}{r}$ averaged over the full disc, keeping the other initial conditions unchanged we obtain:

Disc Type	Language	Value	Difference
Circumprimary	Fortran	21.48627746	4.6×10^{-8}
	Python	21.48627741	
Circumsecondary	Fortran	15.0442992	3.3×10^{-7}
	Python	15.0442995	

Table 3: Comparison of truncation radius values between Fortran and Python for circumprimary and circumsecondary discs using $\frac{H}{r}$ averaged over the full disc.

By comparing the results obtained from Python and Fortran, as shown in Tables 2 and 3, the correctness of the Fortran implementation was verified, ensuring consistency between both approaches.

4 Conclusion

The work done in this lab has calculated the truncation radius of protoplanetary discs in S-type binary star systems.

The parameters b and c were found by first finding the aspect ratio ($\frac{H}{r}$). This was done by finding the mean value of $\frac{H}{r}$ from the results from a population of 1000 simulations of protoplanetary discs around single stars, which were run using the Bern Model of Planet Formation (Emsenhuber et al. 2021). These simulations were performed for three different values of the Shakura-Sunyaev α parameter: 10^{-3}, 10^{-4}, and $5 \cdot 10^{-4}$. This was done for $\frac{H}{r}$ averaged across the full disc, at orbital distances greater than 10 au, and for orbital distances less than 10 au. The $\frac{H}{r}$ values of the 1000 simulations converge starting at 10 au, meaning that when $\frac{H}{r}$ is averaged across the full disc we get a constant value, regardless of the choice of α. This also means that when this is not done a quadratic fit of $\frac{H}{r}$ as a function of α was necessary.

The calculated $\frac{H}{r}$ was then used to calculate the Reynolds number ($\mathcal{R}$), allowing for quadratic polynomial fits to be done for $b(\mathcal{R})$ and $c(\mathcal{R})$ across different values of the mass parameter μ for both the circumprimary and circumsecondary discs. This gave all of the parameters necessary to implement the truncation radius formula.

It was shown that as the masses of the stars get closer in value, symmetry dictates that the truncation radii also become closer in value. Conversely, as the mass of the secondary star decreases, the circumprimary radius increases as it becomes less constrained by the secondary's tidal forces, and the secondary star becomes unable to retain a disc. It was also shown that increasing turbulence (i.e., decreasing viscosity) resulted in a decrease in the radii as the disc becomes more sensitive to gravitational forces. Increasing viscosity results in a radius minima as tidal forces overcome the viscous forces within the disc, but as viscosity keeps increasing the easier outward expansion of material eventually increases the radius.

The implications of these results are significant for planet formation models, as truncation radii limit the spatial extent and mass distribution of protoplanetary discs, influencing planet formation efficiency, migration, and potential habitability. Observed trends align with predictions of material redistribution in truncated discs, laying the groundwork for refining the Bern model for binary systems. Binary systems represent a significant fraction of stellar populations, with approximately half of Sun-like stars residing in multiple star systems (Raghavan et al. 2010). Validating these predictions through observations with ALMA or JWST, particularly in well-characterized systems like Alpha Centauri, could enhance our understanding of how disc truncation shapes planetary systems, offering broader insights into planet formation across diverse astrophysical contexts.

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